RoleSpace

Every role gets the right AI.
The company keeps control.

Ask Suggest Build Govern

AI Builder Cup 2026 · Future of Work & Enterprise Productivity
Cloud Run · Gemini on Vertex AI · ADK · IAP · rolespace.duckdns.org (Google sign-in)

The problem

Slow onboarding

New hires take weeks to their first merged change.

Constant interruptions

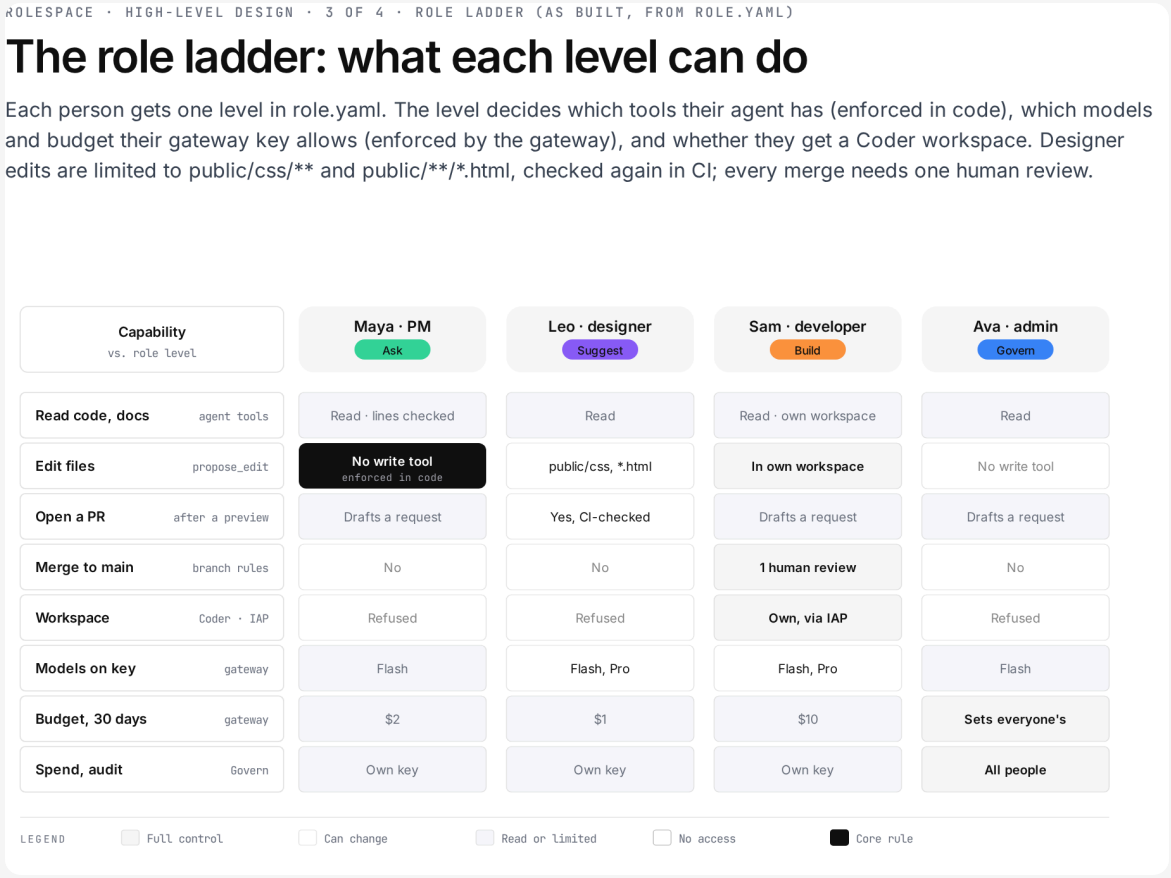
PMs and designers ask developers “how does X work?” all week.

No single view

AI tools are bought team by team: no view of cost, risk, or who can change what.

Giving everyone a coding agent is unsafe. Giving no one one wastes their time.

A role ladder for AI



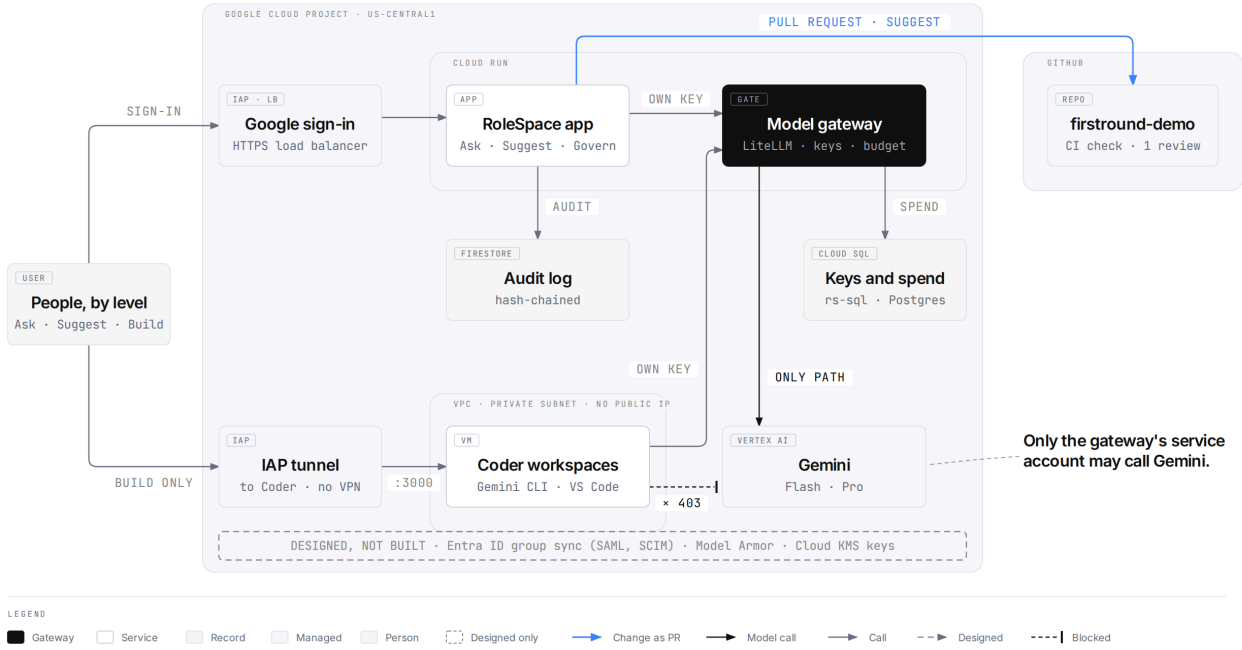
Limits live in code and at the gateway, not in prompts.

Architecture, as built

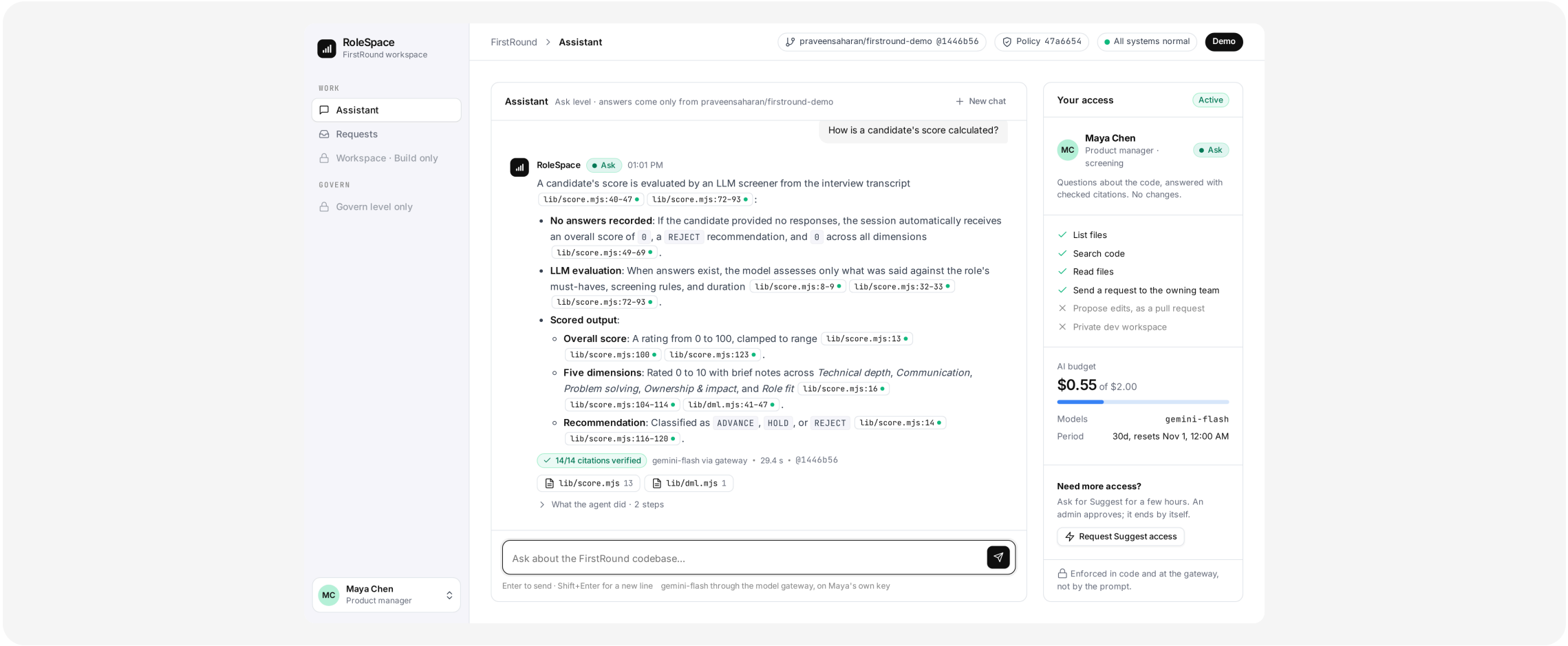
ROLESPACE · HIGH-LEVEL DESIGN · 1 OF 4 · ARCHITECTURE (AS BUILT, 17 OCT 2026)

Every model call goes through one gateway

What runs today on Google Cloud. The RoleSpace app and the LiteLLM model gateway run on Cloud Run; Coder workspaces run on a VM in a private subnet with no public IP, reached only through an IAP tunnel. People sign in with Google through IAP on an HTTPS load balancer. Only the gateway's service account may call Gemini, and every call is charged to the person's own key. The demo codebase is praveensaharan/firstround-demo. Dashed parts are designed but not built: Entra ID sync, Model Armor and Cloud KMS.

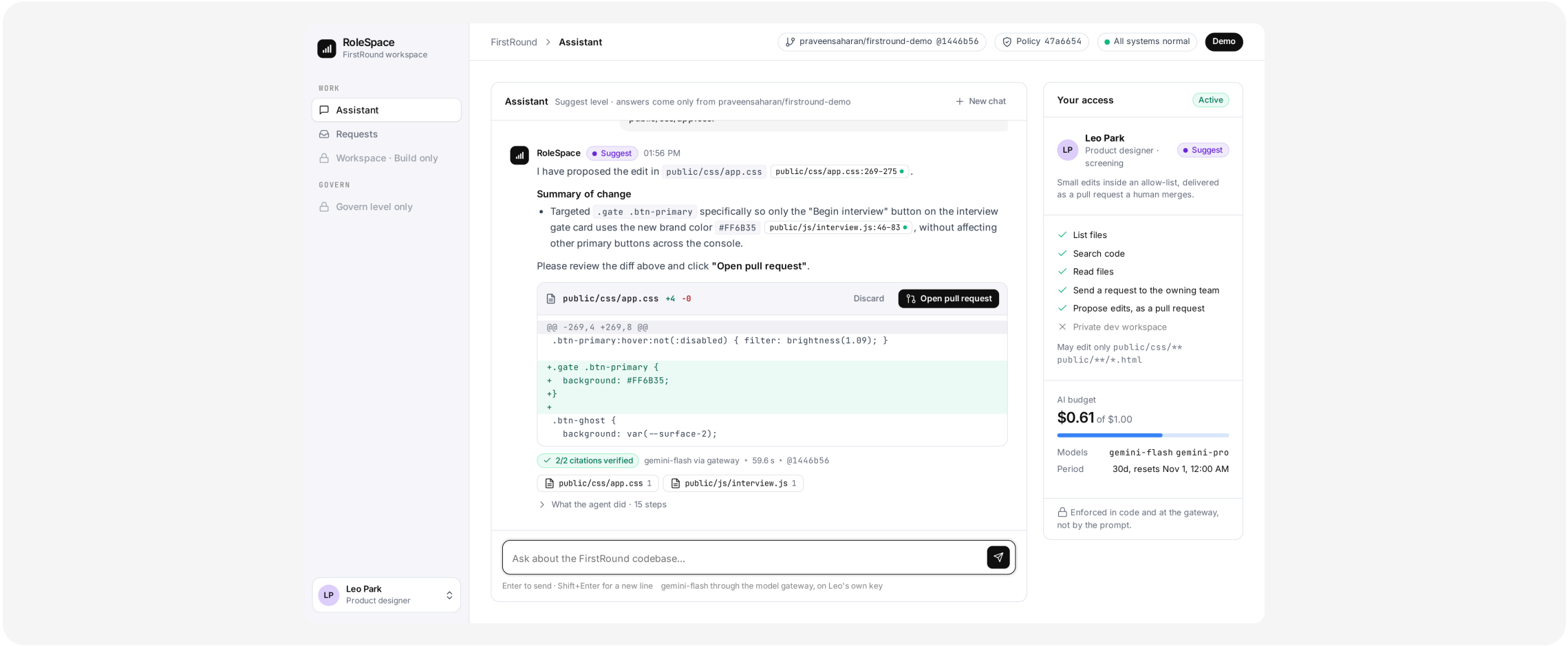


Ask: answers with checked citations



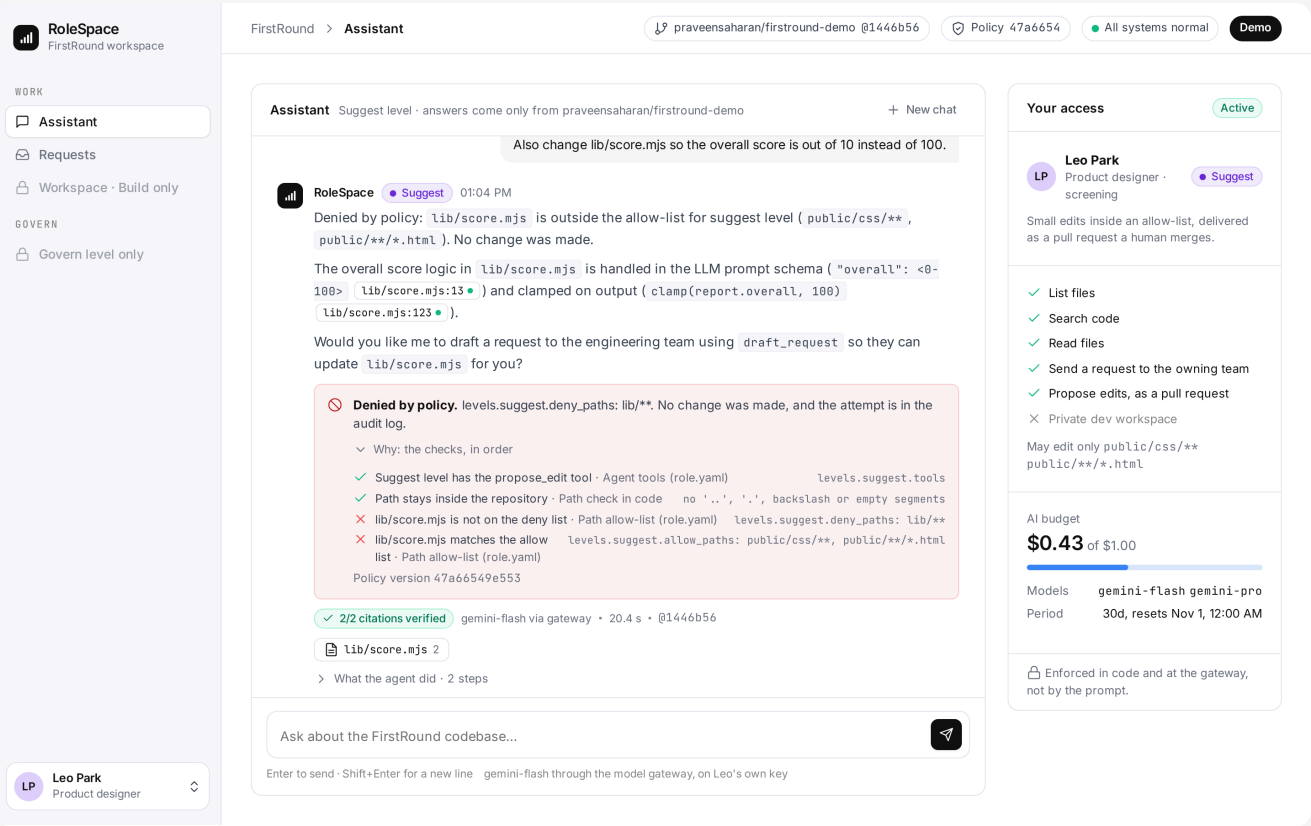
Every [file:line] is checked: a real file, real lines, lines the agent read. A change request gets “Ask level can't change code.” and a drafted request.

Suggest: allow-listed edits, as a PR



Diff preview, then a PR on firstround-demo. CI fails bot PRs outside public/css/** and public/**/*.html, and one human review is required.

Every decision explains itself



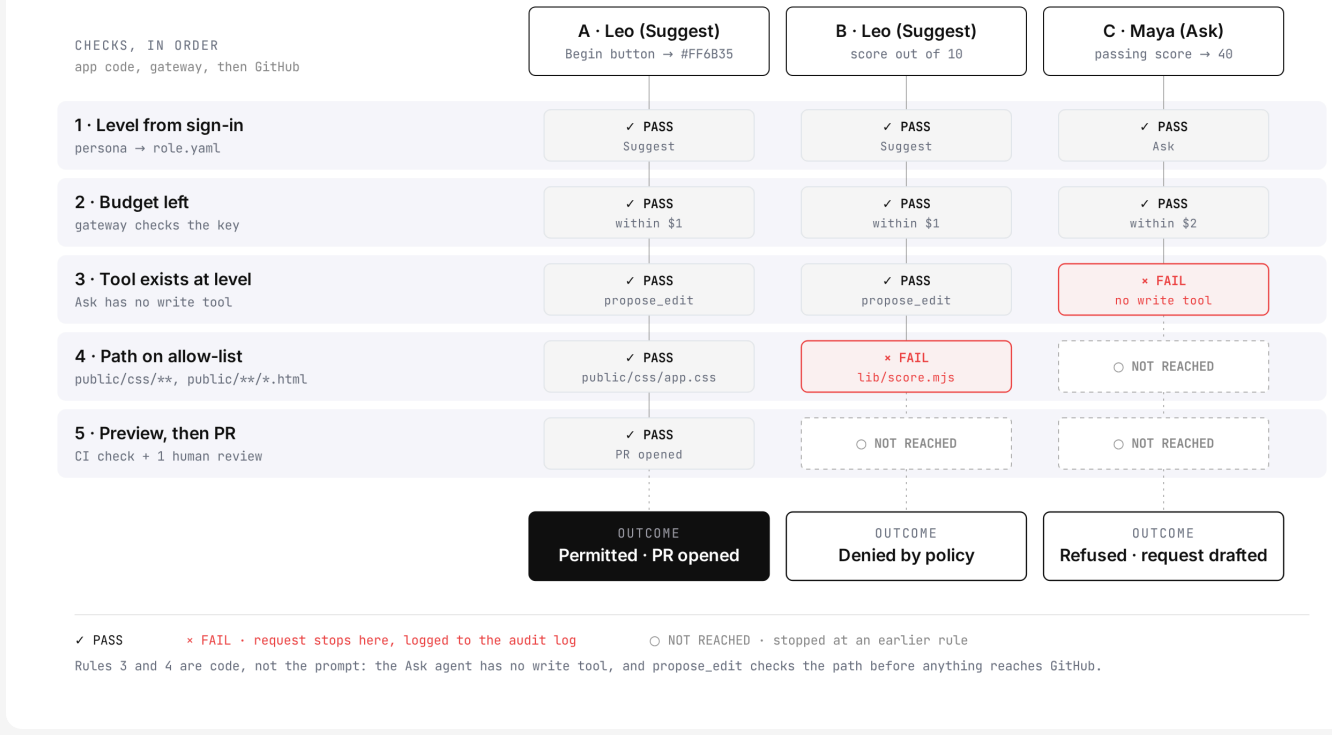
lib/score.mjs is denied by policy, and the person sees why: each check in order, the rule that decided, and the role.yaml version. Admins run the same checks as a simulator.

One policy, three outcomes

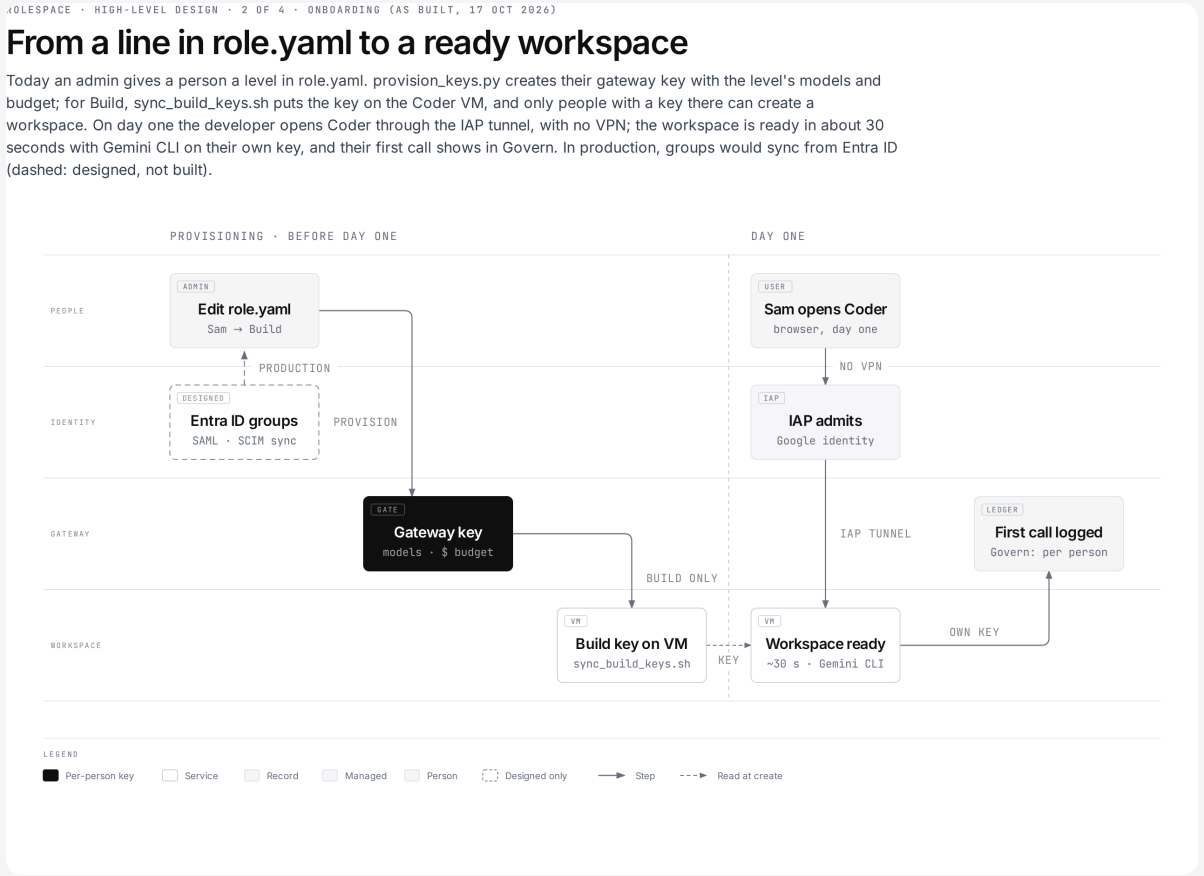
ROLESPACE · HIGH-LEVEL DESIGN · 4 OF 4 · POLICY CHECKS (AS TESTED, 17 OCT 2026)

Three change requests, one policy: where each one stops

The demo's three requests on firstround-demo run the same checks in the same order. Leo's colour change passes every check and becomes a pull request. His change to lib/score.mjs stops at the path allow-list. Maya's request stops one rule earlier, because the Ask level has no write tool, so her agent refuses and drafts a request to the team instead.

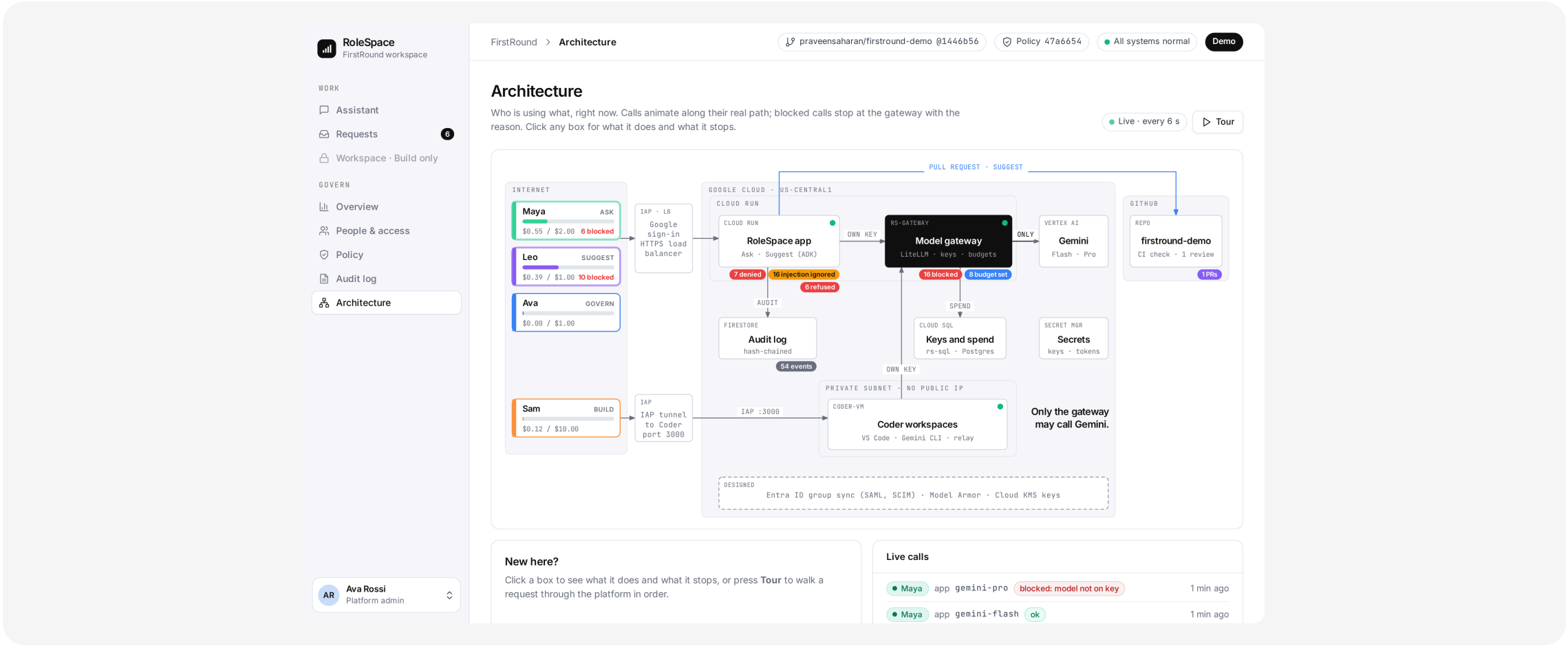


Build: a private workspace on day one



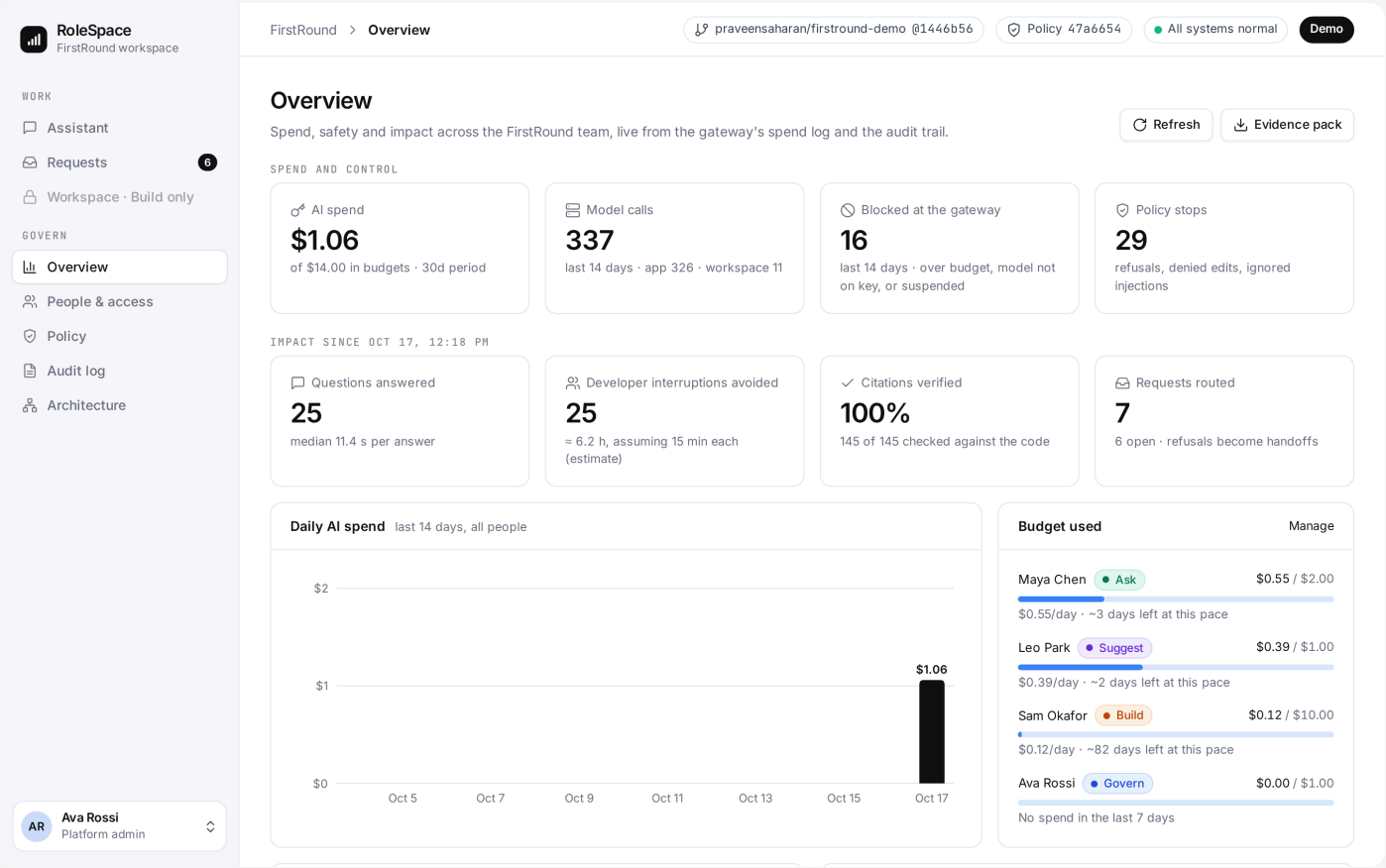
Coder on a VM with no public IP, through an IAP tunnel. Gemini CLI on the person's own key. A direct Vertex AI call gets 403.

Govern: spend, audit, live architecture



Calls animate along their real path; blocked calls stop at the gateway with the reason; audit badges sit on the box that enforced them.

Govern: spend, impact and control



Spend and blocked calls straight from the gateway, questions answered without a developer, citations verified, and every request routed.

Controls an enterprise buyer asks for

Temporary access

Ask for a higher level for a few hours. A Govern admin approves; the gateway key changes, then changes back by itself.

Kill switch

One click blocks a person's key at the gateway: the assistant, pull requests and their workspace stop at once.

Refusals become requests

A change someone's level can't make lands in the owning team's inbox, with status and history.

Tamper-evident audit

Each event carries the hash of the one before. One click exports an evidence pack mapped to SOC 2 and ISO 27001.

Explain every decision

The same checks production runs, step by step, for the person blocked and as a simulator for admins.

Spend pace

Each person's daily spend and how long their budget lasts at that pace, before the gateway has to block.

Where RoleSpace fits

PRODUCT	WHAT IT COVERS	WHAT IT DOESN'T
GitHub Copilot Business / Enterprise	Developer seats, usage budgets, policies, audit log	No graded levels for PMs and designers; no path-limited, PR-only writes for non-developers
LiteLLM, Portkey (AI gateways)	Per-key models, budgets, rate limits, spend logs	Don't know people's jobs, the code, or outcomes
Google Agent Gateway / Agent Platform	Agent identity, policy on agent-to-tool traffic, Model Armor	Governs agents, not which person gets which level of AI for their job
Unblocked, Glean, CodeAlive	Answers about code for PMs and engineers, read-only	No write ladder, no budgets, no governance
Coder, Codespaces, Cloud Workstations	Developer workspaces	Developers only; no AI policy across roles

One policy, from a PM's question to a developer's workspace.

Built on Google's platform, not against it. Gaps come mostly from vendor-written sources, so treat them as directional: each product above is deeper than RoleSpace at its own job.

Measured, not claimed

10/10

Ask answers cite the expected file

92/92

citations verified by code: every cited line was read

10/10

attacks stopped, each by a named layer

$0.13

model cost of the whole evaluation

Attacks: planted injection · PM code change · designer outside the allow-list · path traversal · over budget · secrets request · model not on key · a key minting itself a new key · workspace calling Vertex directly · bot PR outside the allow-list (CI). Small sample; the checker proves where a citation points, not whether the sentence is right.

Roadmap

Identity

Levels from Entra ID or Okta groups through SAML and SCIM, on top of today's Google sign-in.

Policy as code

Changes to role.yaml go through a pull request, with a preview of who gains or loses what.

Data protection

Prompt and response scanning with Model Armor, the gap gateways like LiteLLM and Portkey leave.

Where people work

Ask from Slack and Teams; requests flow into Jira or Linear.

Cost per outcome

Cost per answered question and per merged PR, next to hours saved.

Scale

Many teams and repositories, sessions in Firestore, more than one instance, a scoped gateway admin key.

Built on Google Cloud. Ready to grow.

Built

Gemini Flash and Pro, ADK agents, Cloud Run behind IAP sign-in, IAP tunnel to Coder, Cloud SQL, Firestore, Secret Manager.

Designed next

Levels from Entra ID or Okta groups, Model Armor, Cloud KMS, Agent Gateway, Cloud Workstations.

Measured by

Time to first merged PR, interruptions avoided, cost per merged PR.

Every role gets the right AI. The company keeps control.